

TP : Analyse de Flux et Extraction de Données

1. Préparation du terrain

Nous allons installer un serveur Nginx et utiliser **tcpdump** pour capturer ce qui entre (VM1).

```
ubuntu:~$ sudo apt update && sudo apt install nginx tcpdump curl -y
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
84 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
tcpdump is already the newest version (4.99.4-3ubuntu4.24.04.1).
tcpdump set to manually installed.
The following package was automatically installed and is no longer required:
  squashfs-tools
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  libcurl3t64-gnutls libcurl4t64 nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
The following packages will be upgraded:
  curl libcurl3t64-gnutls libcurl4t64
3 upgraded, 2 newly installed, 0 to remove and 81 not upgraded.
Need to get 1466 kB of archives.
After this operation, 1596 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 curl amd64 8.5.0-2ubuntu10.7 [227 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl4t64 amd64 8.5.0-2ubuntu10.7 [342 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl3t64-gnutls amd64 8.5.0-2ubuntu10.7 [334 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.6 [43.5 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.6 [521 kB]
Fetched 1466 kB in 1s (2853 kB/s)
Preconfiguring packages ...
(Reading database ... 113825 files and directories currently installed.)
Preparing to unpack .../curl_8.5.0-2ubuntu10.7_amd64.deb ...
Unpacking curl (8.5.0-2ubuntu10.7) over (8.5.0-2ubuntu10.6) ...
Preparing to unpack .../libcurl4t64_8.5.0-2ubuntu10.7_amd64.deb ...
Unpacking libcurl4t64:amd64 (8.5.0-2ubuntu10.7) over (8.5.0-2ubuntu10.6) ...
Preparing to unpack .../libcurl3t64-gnutls_8.5.0-2ubuntu10.7_amd64.deb ...
Unpacking libcurl3t64-gnutls:amd64 (8.5.0-2ubuntu10.7) over (8.5.0-2ubuntu10.6) ...
Selecting previously unselected package nginx-common.
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.6_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.6) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.24.0-2ubuntu7.6_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.6) ...
Setting up libcurl4t64:amd64 (8.5.0-2ubuntu10.7) ...
Setting up libcurl3t64-gnutls:amd64 (8.5.0-2ubuntu10.7) ...
Setting up nginx-common (1.24.0-2ubuntu7.6) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
Setting up curl (8.5.0-2ubuntu10.7) ...
```

Lancer une écouteur sur le port 80 (VM1):

```
sudo ss -tulpn | grep :80
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1800 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1558 kB]
Fetched 3484 kB in 2s (2001 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
84 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
tcpdump is already the newest version (4.99.4-3ubuntu4.24.04.1).
tcpdump set to manually installed.
The following package was automatically installed and is no longer required:
  squashfs-tools
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  libcurl3t64-gnutls libcurl4t64 nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
The following packages will be upgraded:
  curl libcurl3t64-gnutls libcurl4t64
3 upgraded, 2 newly installed, 0 to remove and 81 not upgraded.
Need to get 1466 kB of archives.
After this operation, 1596 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 curl amd64 8.5.0-2ubuntu10.7 [227 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl4t64 amd64 8.5.0-2ubuntu10.7 [342 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl3t64-gnutls amd64 8.5.0-2ubuntu10.7 [334 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.6 [43.5 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.6 [521 kB]
Fetched 1466 kB in 1s (1336 kB/s)
Preconfiguring packages ...
(Reading database ... 113825 files and directories currently installed.)
Preparing to unpack .../curl_8.5.0-2ubuntu10.7_amd64.deb ...
Unpacking curl (8.5.0-2ubuntu10.7) over (8.5.0-2ubuntu10.6) ...
Preparing to unpack .../libcurl4t64_8.5.0-2ubuntu10.7_amd64.deb ...
Unpacking libcurl4t64:amd64 (8.5.0-2ubuntu10.7) over (8.5.0-2ubuntu10.6) ...
Preparing to unpack .../libcurl3t64-gnutls_8.5.0-2ubuntu10.7_amd64.deb ...
Unpacking libcurl3t64-gnutls:amd64 (8.5.0-2ubuntu10.7) over (8.5.0-2ubuntu10.6) ...
Selecting previously unselected package nginx-common.
```

résultat:

```
tcp  LISTEN 0      511                0.0.0.0:*        0.0.0.0:*        users:((("nginx",pid=2578,fd=5),("nginx",pid=2575,fd=5))
tcp  LISTEN 0      511                [::]:80          [::]:*          users:((("nginx",pid=2578,fd=6),("nginx",pid=2575,fd=6))
```

vérifie la connexion TCP (VM2):

```
sudo tcpdump -i any port 80 -nn -v
tcpdump: data link type LINUX_SLL2
tcpdump: listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
21:33:22.950587 lo In IP6 (flowlabel 0x320fb, hlim 64, next-header TCP (6) payload length: 40) ::1:33020 > ::1:80: Flags [S], cksum 0x0030 (incorrect -> 0x8dbe), seq 2269685813, win 65476, options [mss 65476,sackOK,TS val 2077856710 ecr 0,nop,wscale 7], length 0
21:33:22.950607 lo In IP6 (flowlabel 0x70084, hlim 64, next-header TCP (6) payload length: 40) ::1:80 > ::1:33020: Flags [S.], cksum 0x0030 (incorrect -> 0x3fbc), seq 2934542195, ack 2269685814, win 65464, options [mss 65476,sackOK,TS val 2077856710 ecr 2077856710,nop,wscale 7], length 0
21:33:22.950617 lo In IP6 (flowlabel 0x320fb, hlim 64, next-header TCP (6) payload length: 32) ::1:33020 > ::1:80: Flags [.], cksum 0x0028 (incorrect -> 0x6852), ack 1, win 512, options [nop,nop,TS val 2077856710 ecr 2077856710], length 0
21:33:22.950671 lo In IP6 (flowlabel 0x320fb, hlim 64, next-header TCP (6) payload length: 228) ::1:33020 > ::1:80: Flags [P.], cksum 0x00ec (incorrect -> 0x5db8), seq 1:197, ack 1, win 512, options [nop,nop,TS val 2077856710 ecr 2077856710], length 196: HTTP, length: 196
POST /login HTTP/1.1
Host: localhost
User-Agent: curl/8.5.0
Accept: */*
X-Auth-Token: Admin123
Content-Length: 25
Content-Type: application/x-www-form-urlencoded

user=root&pass=Hacked2026 [[http]
21:33:22.950675 lo In IP6 (flowlabel 0x70084, hlim 64, next-header TCP (6) payload length: 32) ::1:80 > ::1:33020: Flags [.], cksum 0x0028 (incorrect -> 0x6590), ack 197, win 510, options [nop,nop,TS val 2077856710 ecr 2077856710], length 0
21:33:22.950688 lo In IP6 (flowlabel 0x70084, hlim 64, next-header TCP (6) payload length: 358) ::1:80 > ::1:33020: Flags [P.], cksum 0x016e (incorrect -> 0xbcd8), seq 1:327, ack 197, win 512, options [nop,nop,TS val 2077856710 ecr 2077856710], length 326: HTTP, length: 326
HTTP/1.1 404 Not Found
Server: nginx/1.24.0 (Ubuntu)
Date: Fri, 27 Feb 2026 21:33:22 GMT
Content-Type: text/html
Content-Length: 162
Connection: keep-alive

<html>
<head><title>404 Not Found</title></head>
<body>
<center><h1>404 Not Found</h1></center>
<hr><center>nginx/1.24.0 (Ubuntu)</center>
</body>
</html>
```

```

21:33:22.950879 lo In IP6 (flowlabel 0x320fb, hlim 64, next-header TCP (6) payload length: 32) ::1.33020 > ::1.80: Flags [.], cksum 0x0028 (incorrect -> 0x644a), ack 327, win 510, options [nop,nop,TS val 2077856710 ecr 2077856710], length 0
21:33:22.951338 lo In IP6 (flowlabel 0x320fb, hlim 64, next-header TCP (6) payload length: 32) ::1.33020 > ::1.80: Flags [F.], cksum 0x0028 (incorrect -> 0x6446), seq 197, ack 327, win 512, options [nop,nop,TS val 2077856711 ecr 2077856710], length 0
21:33:22.951362 lo In IP6 (flowlabel 0x70084, hlim 64, next-header TCP (6) payload length: 32) ::1.80 > ::1.33020: Flags [F.], cksum 0x0028 (incorrect -> 0x6444), seq 327, ack 198, win 510, options [nop,nop,TS val 2077856711 ecr 2077856711], length 0
21:33:22.951376 lo In IP6 (flowlabel 0x320fb, hlim 64, next-header TCP (6) payload length: 32) ::1.33020 > ::1.80: Flags [.], cksum 0x0028 (incorrect -> 0x6444), ack 328, win 512, options [nop,nop,TS val 2077856711 ecr 2077856711], length 0

```

Instruction : Observe les drapeaux **[S]** (SYN), **[S.]** (SYN-ACK), et **[.]** (ACK). C'est la preuve que la connexion est établie proprement.

5. Filtrage Avancé (Le couteau suisse)

Un ingénieur ne lit pas des milliers de lignes. Il filtre. Voici les commandes indispensables :

- **Filtrer par IP source uniquement :**

```

ubuntu:~$ curl -H "X-Auth-Token: Admin123" -d "user=root&pass=Hacked2026" http://localhost/login
<html>
<head><title>404 Not Found</title></head>
<body>
<center><h1>404 Not Found</h1></center>
<hr><center>nginx/1.24.0 (Ubuntu)</center>
</body>
</html>
ubuntu:~$

```

6. Le Challenge de l'ingénieur : "L'écoute clandestine"

On va prouver pourquoi le HTTPS est obligatoire. (VM1)

```

sudo tcpdump -i any port 80 -nn -A
tcpdump: data link type LINUX_SLL2
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
21:54:51.839847 lo In IP6 ::1.40340 > ::1.80: Flags [S], seq 3477449254, win 65476, options [mss 65476,sackOK,TS val 2079145599 ecr 0,nop,wscale 7], length 0
..F..@.....P.E..@.....0.....
{..}.....
21:54:51.839867 lo In IP6 ::1.80 > ::1.40340: Flags [S.], seq 1497724166, ack 3477449255, win 65464, options [mss 65476,sackOK,TS val 2079145599 ecr 2079145599,nop,wscale 7], length
'....f@.....P..YEU..E'.....0.....
{..}.....
21:54:51.839878 lo In IP6 ::1.40340 > ::1.80: Flags [..], ack 1, win 512, options [nop,nop,TS val 2079145599 ecr 2079145599], length 0
..F..@.....P.E.'YEU.....(.....
{..}.....
21:54:51.839936 lo In IP6 ::1.40340 > ::1.80: Flags [P.], seq 1:197, ack 1, win 512, options [nop,nop,TS val 2079145599 ecr 2079145599], length 196: HTTP: POST /login HTTP/1.1
..F..@.....P.E.'YEU.....
{..}.....>POST /login HTTP/1.1
Host: localhost
User-Agent: curl/8.5.0
Accept: */*
X-Auth-Token: Admin123
Content-Length: 25
Content-Type: application/x-www-form-urlencoded
user=root&pass=Hacked2026
21:54:51.840943 lo In IP6 ::1.80 > ::1.40340: Flags [..], ack 197, win 510, options [nop,nop,TS val 2079145599 ecr 2079145599], length 0
'....@.....P..YEU..E.....(.....
{..}.....
21:54:51.840113 lo In IP6 ::1.80 > ::1.40340: Flags [P.], seq 1:327, ack 197, win 512, options [nop,nop,TS val 2079145600 ecr 2079145599], length 326: HTTP: HTTP/1.1 404 Not Found
'....f@.....P..YEU..E.....h.....
{..}.....>HTTP/1.1 404 Not Found
Server: nginx/1.24.0 (Ubuntu)
Date: Fri, 27 Feb 2026 21:54:51 GMT
Content-Type: text/html
Content-Length: 162
Connection: keep-alive

<html>
<head><title>404 Not Found</title></head>
<body>
<center><h1>404 Not Found</h1></center>
<hr><center>nginx/1.24.0 (Ubuntu)</center>
</body>
</html>
21:54:51.840125 lo In IP6 ::1.40340 > ::1.80: Flags [..], ack 327, win 510, options [nop,nop,TS val 2079145600 ecr 2079145600], length 0

```

5. Challenge : Diagnostic de Sécurité

Ce test prouve pourquoi le protocole HTTP est proscrit pour les données sensibles.

- **Observation** : Dans ta capture précédente, tu as pu lire `X-Auth-Token: Admin123` et `pass=Hacked2026` en texte clair.
- **Verdict Ingénieur** : Le flux est vulnérable aux attaques de type *Man-in-the-Middle* (MitM). La préconisation immédiate est l'implémentation de **TLS/SSL (Port 443)**.